Danish Syed

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Education

University of Michigan (UoM) — Ann Arbor, USA
Masters in Electrical and Computer Engineering — Sep 2019 - Present
GPA: 3.75/4.00

Indian Institute Of Technology (ISM) — Dhanbad, India
B.Tech in Electronics and Communication Engineering — Jul 2013 - May 2017
GPA: 8.75/10

Research Experience

Architecture and Artificial Intelligence Lab (AR^{2}IL) — Ann Arbor, USA
Research Assistant | Advisor: Prof. Matias del Campo — Jan 2020 - Present

- Co-founded AR^{2}IL to explore the applications of AI in the context of Architecture
- Worked with architecture grad students *(as part of ARCH662)* to answer whether the current state of AI can have a novel sensibility of human creativity at large
- Built a code repository of style transfer and generative methods to empirically test the hypothesis devised by the students [https://arch-ai.org]

Johnson AI Lab, University of Michigan — Ann Arbor, USA
Graduate Student Researcher | Advisor: Prof. Justin Johnson — May 2020 - Present

- Working on efficient conditional 3D scene generation using sparse convolutions
- Developed 3D feature inversion method to understand the features learned by 3D ConvNets and further perform single-shot pointcloud to mesh conversion and vice-versa

Publications

DeepGamble: Towards unlocking real-time player intelligence using multi-layer instance segmentation and attribute detection
Danish Syed*, Naman Gandhi*, Arushi Arora* & Nilesh Kadam
ICMLA 2020 (Full Paper) **[project][arXiv]**

Optimized [AI]rport: Generating Airports Conditioned on Architectural Semantics
Janpreet Singh*, Bruna Iunessanches*, Kylie Williams*, Jun Zhou* & **Danish Syed**
In submission, 2021

Technical skills

- **Languages:** Python, R, C/C++, SQL, MATLAB, Julia
- **Frameworks:** PyTorch, TensorFlow, CUDA, TensorRT, Flask, Django
- **Tools:** Scientific Python Stack, Docker, Slurm, Git, JIRA, Jupyter, LaTeX

Professional Experience

Intvo Inc. — Ann Arbor, USA
Computer Vision Engineer Intern — Jun 2020 – Aug 2020

- Built a novel Convolution Neural Network and augmented it with Kalman Filtering to track pedestrians and predict their behaviour in the context of Autonomous Vehicles

* Indicates equal contribution

- Deployed the model using PyTorch and TensorRT to test it in the city of Ann Arbor

ZS Associates — Pune, India
Data Scientist — Jul 2017 – May 2019

- Designed a video recognition system *(DeepGamble)* for a gaming client to create a digital understanding of blackjack players, which proposed over $100M of annual savings
- Performed Monte-Carlo simulations for a gaming client to understand the customer journey and their gameplay behavior to effectively tailor the promotion techniques
- Implemented an LSTM autoencoder for a manufacturing client to analyze the sensor data and predict anomalies in their batch and suggested key-drivers for non-conformance
- Identified Key Opinion Leaders in the health-care domain from Twitter data by leveraging page-rank algorithm

Relevant Projects

Benchmark_VO *(EECS 568: Mobile Robotics Course Project)*
- Evaluated various detector-descriptor combinations used in Visual Odometry to explore the effect of semantic understanding in localization for `SE(3)` poses
- Compared Absolute and Relative Pose Errors along with runtime on KITTI dataset

SICGAN *(EECS 504: Computer Vision Course Project)*
- Built a conditional GAN framework for generating 3D meshes from a single image
- Achieved qualitatively better 3D reconstructions as compared to the baseline

MedLens *(ZS Quest Hackathon Winners)*
- Designed a web app using Django and bootstrap to help health-care researchers with factual question answering based on uploaded documents or articles from PubMed
- Implemented a Bi-Directional Attention Flow network to perform question-answering

Kaggle Projects *(Silver/Bronze Medals)*
- Built a ConvNet to identify atmospheric conditions and land type from satellite images
- Experimented with U-Nets to count different types of sea lions in aerial photographs
- Built an ensemble of LSTM and XGBoost models to predict similar questions on Quora

Teaching and Mentoring

GSI, EECS 442: Computer Vision (CV), University of Michigan	Winter 2021
GSI, EECS 598: Deep Learning for CV, University of Michigan	Fall 2020
TA, ARCH 662: Architecture and AI, University of Michigan	Winter 2020
Instructor, Architecture and AI Workshop, DigitalFutures	Winter 2020

Leadership

- Managed the **TEDx IIT Dhanbad** event as the Executive Director
- Taught mathematics to underprivileged Class V kids at **Kartavya**, a student run NGO
- Managed the **Robotics Club, IIT Dhanbad** as the Club Head for two years